

CoRoT-2b

R-0575

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_230731 · created 2026-07-24T16:44:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460156.7499 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.065
Transit depth (Rp/Rs)^2	4.65 +/- 2.82 [%]
Orbital Inclination (inc)	88.75 +/- 2.04
Airmass coefficient 1 (a1)	0.34 +/- 0.21
Airmass coefficient 2 (a2)	0.76 +/- 1.26
Scatter in the residuals of the lightcurve fit is	63.32 %
Best Comparison Star	None
Optimal Aperture	6.01
Optimal Annulus	13.55
Transit Duration (day)	0.0987 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.065 vs 0.1667 (deviation 0.74σ)
Duration fit vs published	0.0987 d vs 0.095 d (deviation 0.55σ)
Mid-time O–C	-6.3 min
Depth SNR	3.2
Residual scatter	63.32 %

Light curve

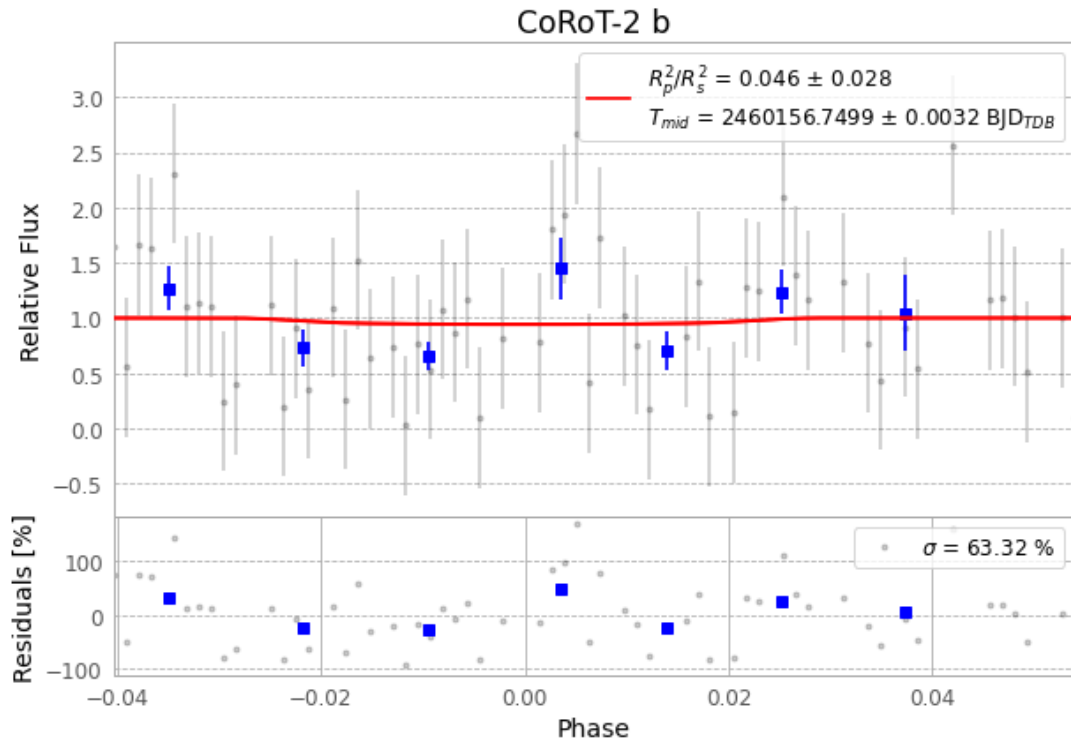


Figure 1 — FinalLightCurve_CoRoT-2 b_2023-07-30.png (SHA-256 2b9a43fb0cc7cb0a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c38eb0f6e2eee7e5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

78 FITS frames (51 MB), CoRoT-2230731040915.FITS → CoRoT-2230731080615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-230731.log (651cd6356717...), FinalLightCurve_CoRoT-2 b_2023-07-30.png (2b9a43fb0cc7...), FinalLightCurve_CoRoT-2 b_2023-07-30.csv (f30edc1d144f...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 16:44Z.